



Self-Interruptions of Non-Driving Related Tasks in Automated Vehicles: Mobile vs Head-Up Display

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ABSTRACT

Automated driving raises new human factors challenges. There is a paradox that allows drivers to perform non-driving related tasks (NDRTs), while benefiting from a driver who regularly attends to the driving task. Systems that aim to better manage a driver's attention, encouraging task switching and interleaving, may help address this paradox. However, a better understanding of how drivers self-interrupt while engaging in NDRTs is required to inform such systems. This paper presents a counterbalanced within-subject simulator study with N=42 participants experiencing automated driving in a familiar driving environment. Participants chose a TV show to watch on a HUD and mobile display during two 15min drives on the same route. Eye and head tracking data revealed more self-interruptions in the HUD condition, suggesting a higher likelihood of a higher situation awareness. Our results may benefit the design of future attention management systems by informing the visual and temporal integration of the driving and non-driving related task.

Author Keywords

Conditionally Automated Vehicles; Self-Interruption; Attention Management; Non-driving related Task; Task Engagement; Human-automation Interaction

CCS Concepts

•Human-centered computing → HCI theory, concepts and models; User studies; Laboratory experiments;

INTRODUCTION

Automated Vehicles (AVs) are promising to reduce crashes in the future. However, the development towards fully automated vehicles faces new human factor challenges. It turns decades of research in the context of driver distraction upside down [25], and raises new questions around the relationship between the driving task and any non-driving related task (NDRT), the

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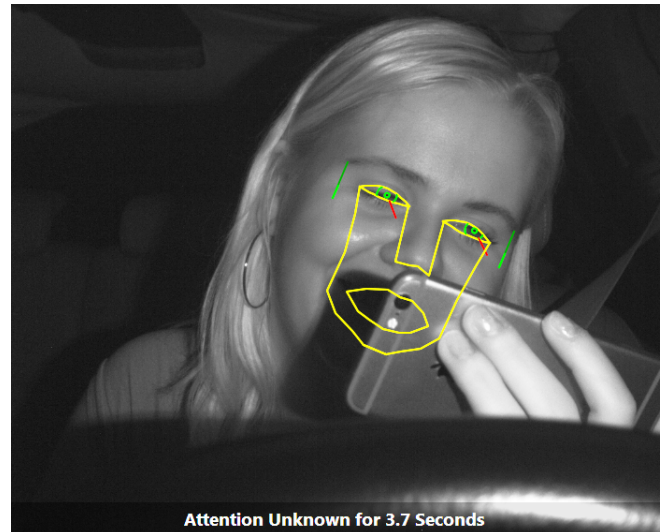


Figure 1. Driver Monitoring System's view of the participant in the mobile display condition.

notions of primary and secondary task, as well as the switching between and interruption of those tasks.

In manual driving (SAE Level 0-2 [33]) the drivers perform a primary task, the driving task, and, unfortunately, distracting secondary tasks, such as interactions with the in-vehicle radio, making a phone call, or talking to passengers. These distractions cause a lack of awareness of the driving situation [9] and increase the delays for critical responses [8] with potentially fatal consequences.

Increased vehicle automation flips those existing notions of primary and secondary tasks on their head, at least during automated driving. Conditional and higher automation (SAE Level 3 and above [33]) promises to free drivers from the driving task, and enables them to engage in NDRTs instead (see Figure 1, such as eating, gaming, or watching a movie [27]). This promise is highly anticipated from a consumer's perspective [30], and the likelihood that drivers engage in such NDRTs is significantly higher with increasing levels of automation [28, 23, 14].

Human factors challenges arise in automation levels that demand (SAE Level 3) or benefit (SAE Level 4) from a driver who is able to take control of the vehicle safely under cer-

tain system limit situations. E.g., for situations when system limitations are encountered in SAE L3, the driver must remain fallback-ready, a state vaguely described as being able to safely control the vehicle "at least several seconds after providing the fallback-ready user with a request to intervene" [33]. These situations create complex stages of transition of control, each with a different understanding of primary and secondary task, and a different understanding around task interruptions, multi-tasking, and interleaving of tasks.

This paper builds on the task interleaving framework of Janssen et al. [17], and focuses on the initial stages within this framework, during which the driver engages in non-driving tasks (stage 0 of the framework [17]) and chooses to "self-interrupt" (stage 1). Such self-interruptions are defined as discretionary task switches due to one's own choice [17]. It is assumed that a higher number of longer self-interruptions towards the driving task is more likely to increase situation awareness (SA) because *perception* is the required first stage followed by *comprehension* and *projection* in Endsley's SA model [9]. And higher SA may improve going through subsequent stages of the control transition in terms of time and quality [17, 34]. As such, the overall transition of control may benefit from a system that manages interruptions in an optimised way [28, 24].

This research aims to contribute to the CHI community by providing insight into the temporal relationship between the visual attention towards an engaging NDRT during automated driving, and its self-interruptions. A better understanding of how self-interruptions currently occur during automated driving may inform future attention or interruption management systems. We explore this by comparing two display devices used to engage in NDRTs in cars: head up display (HUD) and mobile devices. As such, the research questions this paper aims to answer is: **RQ.** *How do different display devices (HUD vs mobile) affect a) long NDRT fixations in the NDRT, and b) self-interruption behaviour, during automated driving?*

RELATED WORK

Interruptions are intertwined in multitasking. In office environments, interruptions may "just" affect productivity [11, 6, 22, 18]. In safety-critical environments, such as aviation, healthcare, and driving (see overview by Janssen et al. [16]) they can be outright dangerous. In manual driving, primary driving task interruptions are commonly referred to as distractions, a widely studied and increasing road safety issue, in particular, those caused by mobile phone use in the car. Although distraction from the driving task is less of an issue in automated driving such as SAE Level 3/4, interruptions from NDRTs towards the driving task may help to keep the driver fallback-ready [37, 17, 29, 34].

Interruptions are typically differentiated between a) exogenous (external alert, forced interruption) or b) endogenous (self-interruption) initiated. A forced interruption of a task in response to an external alert can be stressful, can lead to bad performance and user experience, and reduces the chances of a more seamless task resumption of the previous task [7, 12]. To counteract these effects, previous work aimed to identify or predict good or opportune moments to interrupt users/drivers,

e.g. while they are using their mobile phone [31] or while they drive a manual car [35]. In the automated driving context, pre-warnings show to reduce preparation costs for takeovers [36, 4].

Compared to exogenous interruptions, self-initiated (or discretionary) interruptions are natural and not intrusive since they are based on the user's *own* choice or priorities [28, 17]. During automated driving, a fallback-driver might use self-interruptions to switch between an entertaining NDRT and supervision tasks to remain situation-aware or fallback-ready [24, 34, 5], also referred to as task-interleaving [17]. This kind of task-interleaving has the benefit to reduce stress and may help with resuming the original task [28, 17].

Self-interruptions are influenced by different factors. Positive factors include reorganisation, exploration, stimulation, whereas negative ones include obstruction, exhaustion or frustration [1, 2]. People also tend to interleave at natural breakpoints [15], whichever those may be in different contexts or tasks. In automated driving, the NDRTs and the frequency of a self-interruption can be an indicator of some awareness about the driving situation [17] and might lead to further self-interruptions. In this paper, we focus on how the choice of the display device influences ones self-interruptions.

In terms of in-vehicle display devices that provide a source of interruption while driving, personal display devices such as mobile phones and fully integrated head-up or windscreen displays have both received a lot of attention in the automotive and road safety literature. Mobile phones have been studied pre-dominantly as a source of distraction during manual driving, taking a driver's eyes off the road. Head-up displays (HUDs) adopt techniques to project information in the proximity of the central visual field as a way to limit detrimental effects of off-road glances, e.g. when displaying vehicle navigation information. Many studies show HUDs are well-accepted and improve driver's performance in the manual driving context [21, 20, 19].

In automated driving when NDRTs are allowed for drivers, studies on the application of HUD [13] and implications of mobile phone use are limited [4]. It is assumed that using a HUD to present NDRTs will provide more benefits over conventional devices such as a mobile phones in terms of increasing driver's visual attention on the road. This assumption is grounded in Wicken's SEEV Model of Visual Attention Allocation [39, 38], in particular the display devices' difference in terms of salience and effort in this use case. However, to identify this, it is necessary to investigate driver's self-regulation activities when they are engaging in NDRTs by using a HUD and a conventional handheld device respectively.

METHODOLOGY

Apparatus

The experiment was conducted using an Advanced Driving Simulator. It consists of a real vehicle cabin (Holden Commodore) with automatic transmission, a surround-sound system for engine and environmental noise, and a 6 degree of freedom (DOF) motion platform that can move and twist in three dimensions. Drivers in the simulator have a 180° field of

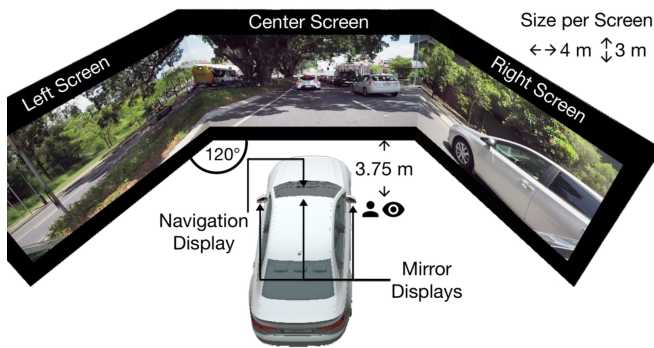


Figure 2. 3D sketch of the simulator setup [10]

view (FOV) provided by three front-view projectors around the vehicle. Wing mirrors and the rear-view mirror are replaced by LCD monitors to simulate the side and rear-view images, suggesting a 360° surround view when looking forward.

To enhance the participants' immersion, sense of reality and familiarity in an automated driving environment, the study used videos recorded in real-world driving as the experiment scenario [10], rather than the simulator simulation software, which has limitations. Our video setup allows participants to better relate to the drive. The scenario videos were composed of seven videos recorded from real-world driving: three at 60° out the front to create a forward 180° view when projected on the three front projection screens with near seamless stitch, three to represent three different rear mirror views played back in the simulators LCD screens (see Figure 2), and one screen-capture video of the navigation played back in the in-vehicle navigation HMI. Driving sounds were recorded through the in-vehicle cameras and also played back in the simulation. The seven videos and sounds were integrated, stitched and synchronised using a custom tool. Manual driving following a Take Over Request (TOR) is simulated by asking participants to "shadow drive", i.e., replicate the car's lateral and longitudinal movements and respond to any hazards. The unique setup and the two pilot studies, which had been conducted prior to the study to iterate and improve immersion and the shadow drive concept, are extensively discussed in previous work [10].

The Driver Monitoring System (DMS) developed by Seeing Machines Limited was used to collect drivers' eye-movement data. It uses a camera installed on the front dashboard of the vehicle (above the steering wheel and instrument cluster for a least obstructed view) to monitor drivers' head and eye movement (see Figure 1). The DMS is representative of such systems being installed in some production cars today and those being made mandatory in European vehicles by the EU as part of revised General Safety Regulation¹. The recorded videos were processed by Seeing Machines' algorithms to output the detailed position, movement, orientation information of the head and eye gaze relative to the in-vehicle and simulator space.

¹Press-release of the European Commission https://europa.eu/rapid/press-release_IP-19-1793_en.htm, March 2019, Retrieved: September 2019

Driving Scenario

To record the driving scenarios used in this study, we drove a car manually in the city where the study took place on one of the most recognisable routes. To mimic the performance of an automated vehicle, we followed the speed limits, drove passively in terms of following distances, and drove highly anticipatory in terms of lane selections. All trips were approximately 15 min long, on the same route covering inner-city roads (first ~5 min), a major highway (~8 min) and suburban roads (last ~2 min), during the same time of the day with similar traffic density. A TOR was issued when exiting the highway in all three scenarios, akin to the vehicle exiting its *Operational Design Domain* [33].

Participants

Forty-six participants attended the experiment, including 23 females and 23 males. They aged between 20-62 years, with an average age of 32 years old and standard deviation of 8.4 years. All participants held a valid Australian driver's licence, the country the study was conducted. They were recruited from local communities who were familiar with local roads around the city.

Experiment Design and Procedure

The study used watching self-selected TV shows as the NDRT, to be watched on two different display devices: a) In the *mobile* condition, a smartphone (iPhone 6) that drivers needed to hold in their hands, and b) in the *HUD* condition, a head-up-display that was simulated by overlaying it on the front projection screen with 61.5% transparency, 76cm × 43cm in size, at approx. 3.75m from the driver's eye. Self-selection of the video content was used so that participants would voluntarily and naturally [27] engage in a TV show they enjoy watching, as they might while driving by themselves in the real-world.

The study used a within-subject design. The within-subject factor were three treatment conditions (two video displays (mobile and HUD respectively) and one baseline (no video display)). The conditions and the three similar driving scenarios were counterbalanced among all participants to reduce order and learning effect. As this paper focuses on NDRT self-interruptions, only the two video display conditions with an NDRT are analysed.

Upon arrival, each participant signed a consent form to participate and was informed that their participation was entirely voluntary. They were introduced to the experiment procedure (e.g. that they could quit the experiment at any time in case of motion sickness or any discomfort) and what they would experience during the experiment, including the NDRT and the automated driving (with the instruction that if the vehicle detected a system failure or a situation beyond its own capabilities, it would ask to take over control).

All participants could self-select two episodes of a TV show to watch during the two drives. Before the formal drives, all participants completed a familiarization drive in the simulator to experience the automated driving and practice shadow driving (approx. 5 min), followed by a motion sickness check to assess their suitability to continue the study (standard simulator

procedure). After completing the experiment, each participant received an AUD 70 gift voucher as a compensation for their participation. The study was conducted in accordance with the national code for the Responsible Conduct of Research and approved by the university's ethics board and the Australian Code for the Responsible Conduct of Research (QUT Ethic approval number 1700000425).

Data Collection and Analysis

During the experiment, drivers' head and eye-movement data were recorded at a sampling rate of 60 Hz. Key variables were extracted from the raw data within the same 5 minutes section of each drive in the middle of the "uneventful" automated driving on the highway (so the section from ~7-12 min of each entire drive). In addition, demographic data including age and gender were collected. Within the scope of this paper, we investigated the differences in self-interruption behaviour between consuming a TV show over a HUD vs a mobile device, therefore the data from the baseline drive is omitted from the analysis.

Unlike in the HUD condition, we were not able to use the DMS data to accurately determine a participant's fixations on the mobile phone because the drivers were allowed to use the phone at any position they felt comfortable. Consequently, this resulted in two different approaches for each condition to determine whether a participant watched the TV show (NDRT), interrupted it (self-interruption), or whether it was unknown (also see Limitations section). The data of the mobile condition were therefore manually annotated frame by frame using the three labels, "NDRT", "Self-interruption" and "Unknown". In this study, if drivers' gaze focused on the HUD or mobile phone for over two seconds, it was then defined as a long NDRT fixation. Likewise, if drivers' gaze shifted away from the HUD or mobile phone for over two seconds, it was defined as a self-interruption. Since there is no consensus yet around what duration to use, we decided to use two seconds as this represents the typical duration used in the visual driver distraction literature.

To answer the research questions in this paper, the labels "NDRT" and "Self-interruption" within the same 5 minutes section of each drive were used to extract four variables for statistical analysis. The four variables for HUD and mobile display were (1) the number of long NDRT fixations, (2) the duration of long NDRT fixations, (3) the number of self-interruptions from the NDRT and (4) the duration of the self-interruptions.

The extracted variables include both count data and continuous data. Thus, different models were used to statistically analyse the variables. For the count data, Poisson regression was used to explain the variation in the number of long NDRT fixations and the number of self-interruptions in terms of age, gender and video display conditions. For the continuous data, mixed-effect Gamma regression was used to explain the variability in the fixation duration and self-interruption duration. A significance level of 0.05 was applied on both regression models.

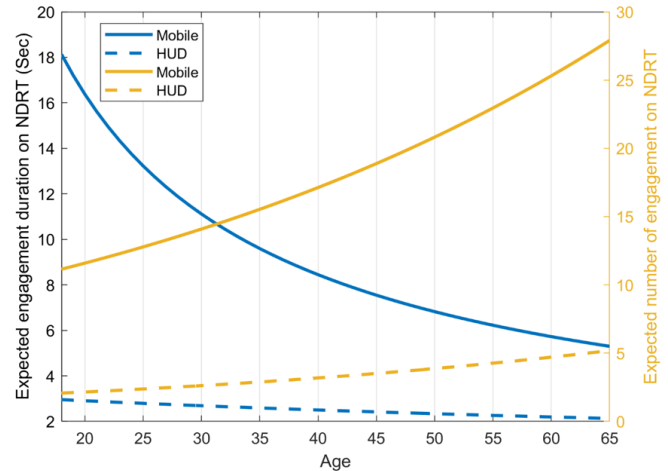


Figure 3. Number of Long NDRT Fixations

RESULTS

Long NDRT Fixations

Number of Long NDRT Fixations

The number of long NDRT fixations describes the total number of times that drivers focus on the self-selected TV show for more than two seconds. Table 1 (top) and Table 3 lists the descriptive statistics and Poisson regression model results of the number of long fixations. The model results show that number of engagements in the NDRT was significantly influenced by age and the display condition. As shown in Figure 3, as the drivers' age increased, the number of long NDRT fixations also increased. Comparing the HUD and mobile phone conditions, drivers had a larger number of long NDRT fixations when they watched videos through a mobile phone than HUD.

Duration of Long NDRT Fixations

The duration of long NDRT fixations describes the average duration of each time that drivers fixated in the entertainment video within the 5 min automated driving for more than two seconds. The descriptive statistics of long NDRT fixation duration is listed in Table 3. The Gamma regression (mixed effect) model results are shown in Table 1 (bottom). Both, age and video display condition, had significant effects on drivers' fixation duration on NDRT. Overall, drivers' fixation duration on NDRT decreased as their age increased (see Figure 3). Drivers watching videos through a mobile phone had longer fixation duration than drivers watching videos through the HUD (see Figure 3).

Self-Interruption from NDRT

Number of Self-Interruptions

The descriptive statistics of number of self-interruptions are listed in Table 4. The Poisson regression model results are shown in Table 2 (top). Video display condition was the only factor that had a significant effect on the number of self-interruptions. The drivers in the HUD condition had more self-interruptions than drivers in the mobile phone condition.

	<i>Coefficient estimate</i>	<i>SE</i>	<i>tStat</i>	<i>DF</i>	<i>pValue</i>	<i>Lower</i>	<i>Upper</i>
<i>Intercept</i>	2.0375	0.301	6.7691	84	1.6412e-09	1.4389	2.636
<i>Age</i>	0.019521	0.008796	2.2193	84	0.029165	0.0020288	0.037012
<i>Gender</i>	0.044445	0.14859	0.29912	84	0.76559	-0.25104	0.33993
<i>Display Condition</i>	-1.6827	0.095229	-17.67	84	4.957e-30	-1.8721	-1.4933

	<i>Coefficient estimate</i>	<i>SE</i>	<i>tStat</i>	<i>DF</i>	<i>pValue</i>	<i>Lower</i>	<i>Upper</i>
<i>Intercept</i>	0.00040218	0.00064147	0.62697	67	0.53281	-0.0008782	0.0016826
<i>Age</i>	0.000046393	0.000019138	2.4241	67	0.018052	0.0000081928	0.000084593
<i>Gender</i>	-0.00050606	0.00031328	-1.6153	67	0.11094	-0.0011314	0.00011926
<i>Display Condition</i>	0.0046684	0.00027772	16.81	67	9.9412e-26	0.0041141	0.0052227

Table 1. Poisson regression model with number of long NDRT fixations (top) and mixed Gamma regression of duration of long NDRT fixations (bottom)

	<i>Coefficient estimate</i>	<i>SE</i>	<i>tStat</i>	<i>DF</i>	<i>pValue</i>	<i>Lower</i>	<i>Upper</i>
<i>Intercept</i>	2.3007	0.19209	11.977	84	7.3405e-20	1.9187	2.6827
<i>Age</i>	0.0044945	0.0055673	0.80731	84	0.42177	-0.0065766	0.015566
<i>Gender</i>	-0.16962	0.093026	-1.8233	84	0.071812	-0.35461	0.015376
<i>Display Condition</i>	0.75675	0.056062	13.499	84	9.5247e-23	0.64526	0.86823

	<i>Coefficient estimate</i>	<i>SE</i>	<i>tStat</i>	<i>DF</i>	<i>pValue</i>	<i>Lower</i>	<i>Upper</i>
<i>Intercept</i>	0.0033041	0.00085974	3.8431	83	0.00023698	0.0015941	0.0050141
<i>Age</i>	-0.000018013	0.000025373	-0.70995	83	0.47972	-0.000068479	0.000032452
<i>Gender</i>	0.00058254	0.00041435	1.4059	83	0.16349	-0.00024159	0.0014067
<i>Display Condition</i>	0.0013867	0.00026868	5.1612	83	0.0000016481	0.00085233	0.0019211

Table 2. Poisson regression model of number of self-interruptions (top) and mixed Gamma regression of duration of self-interruptions (bottom)

Long NDRT fixations (number of):			
<i>Display Condition</i>	<i>Mean</i>	<i>SD</i>	<i>Effect size *</i>
Mobile	16.311	8.906	1.8871
HUD	3.000	4.493	
Long NDRT fixation duration (in seconds):			
Mobile	18.575	18.614	1.201
HUD	2.767	0.480	

Table 3. Number and duration of long NDRT fixations

Self-interruptions (number of):			
<i>Display condition</i>	<i>Mean</i>	<i>SD</i>	<i>Effect size *</i>
Mobile	10.978	7.967	-1.605
HUD	23.047	7.041	
Self-interruption duration (in seconds):			
Mobile	6.277	5.361	0.594
HUD	3.915	1.698	

$$* \text{ Effect size} = \frac{\mu_1 - \mu_2}{SD_{\text{pooled}}} \quad SD_{\text{pooled}} = \sqrt{\frac{SD_1^2 + SD_2^2}{2}}$$

Table 4. Number and duration of self-interruptions

Duration of Self-interruptions

Table 4 shows the descriptive statistics of self-interruption duration. Table 2 (bottom) lists the Gamma regression (mixed effect) model results of self-interruption duration. Similarly, the duration of self-interruption is significantly affected by

video display conditions. Compared with the HUD condition, the duration of self-interruption was significantly longer in the mobile phone condition.

DISCUSSION

In automated driving, especially in SAE Level 3, drivers' interruptions from NDRT activities can have critical safety benefits. Interruptions from NDRTs help increase drivers' awareness of the surrounding environment, which may contribute towards keeping the driver fallback-ready. Attention management systems that better manage such interruptions may be required. To that end, our findings may aid their development, acceptance and usability.

Self-interruptions are self-initiated task-switches that occur at the driver's own discretion. The HUD showed to facilitate more self-interruptions when drivers were engaged in the NDRT. In other words, when using a HUD, drivers' discretion seems to be influenced by the driving related information that is visually appearing closer to the NDRT. In a way, the immediate surrounding environment "distracts" drivers from the NDRT, which in this case could be positive as it suggests a potentially higher situation awareness.

In terms of visual attention allocation (see SEEV [39, 38]), using the HUD, the *effort* involved in retrieving environmental information is lower, while the *salience* of the environmental information is higher, whereas the *salience* of the TV show is higher in the mobile condition, where it is contrasted against

the dark cockpit. Further building on Wicken's SEEV model, we note that *expectancy* and *value* of the TV show and driving scene were somewhat controlled for between conditions by using similar driving scenes and counterbalancing them, and participants watching different parts of the same self-selected TV show.

While the NDRT's self-interruption is based on the driver's discretion, so is coming back to the NDRT, i.e., interrupting the self-interruption, or the task interleaving. Our results show that handheld, mobile device displays lead to longer duration of self-interruptions. While a small portion of this longer duration can be attributed to the added *effort* in head movements (up and down) for some participants, it does suggest that it also took drivers longer to collect a satisfying amount of information from the surrounding environment to come back to the NDRT at their own discretion. The HUD seems to facilitate the task interleaving more naturally, fluently, and efficiently.

The number of NDRT long NDRT fixations and longer duration of NDRT fixations in the mobile condition also suggests that drivers were more likely to engage more deeply in the NDRT compared to the HUD. In a way, participants seem to "forget" about the surroundings when consuming the TV show on the handheld display, which, if focused on in a head down position, blocks out a environmental information in the peripheral.

Regarding drivers' characteristics, the study shows that younger drivers have longer NDRT fixation durations. In contrast, older drivers seemed to interleave tasks more often as the number of long NDRT fixations and number of self-interruptions increased with age (in this study though the age effect is not significant). Generally, driving experience increases with drivers' age. More experienced drivers tend to be more vigilant at the road and traffic environment and they are more capable than less experienced drivers in distributing attention among multiple tasks [26, 3]. This may have implications for future attention management systems that aim to keep drivers fallback-ready, because it suggests that younger drivers' natural discretionary behavior is further away from the target behaviour of more task interleaving.

Overall, while HUDs have continuously demonstrated benefits in manual driving [21, 20, 19], our paper demonstrates benefits for automated driving as well (compared to mobile). The visual integration of the driving and NDRT aids self-interruption behaviour and task interleaving in such a way that more attention is voluntarily attributed to the driving task, as hypothesised by Schroeter et al. [34]. In addition, our study showed that the HUD has a benefit in that real-world driver monitoring systems, such as the one used in this study, are able to better assess the state of the driver when engaging content is provided within the field of view of the windscreen.

Our paper provides a greater understanding of how people adapt their natural interruption and interleaving behaviour, which may inform the design of attention management systems to reduce demand on the driver [5]. In future work, we aim to explore subtle visual attention requests around the HUD to

nudge the frequency of self-interruptions in such a way that they are still perceived as discretionary.

LIMITATIONS

Using two different approaches to label the data for each condition may have affected subsequent results. However, given the nature of each condition, this was necessary. We wanted to provide participants with the freedom to behave naturally in both conditions, therefore, we did not give them instructions about how to sit, what *to do* or *not to do*, and where to put or how to hold the mobile phone. The mobile phone condition therefore resulted in more variation and a more challenging default behaviour (looking down), ultimately leading to more unknown data points that needed to be resolved manually. In the HUD condition, where NDRT and the self-interruption (e.g. to check traffic) may be in close proximity, it is impossible to distinguish between the two by manually reviewing the video of the driver. For this, relying on the DMS' processing and ability to detect such smaller changes is the only option in our setup.

The HUD was simulated through projection overlay on the driving simulation screen, eliminating typical HUD characteristics such as eye-box, field of view, and most notably, depth perception. The NDRT was perceived at the same depth plane as the driving scene, and participants did not have to accommodate to varied distances between them. This may have reduced the effort and increased salience to self-interrupt compared to HUDs on the market today.

Motion has been shown to influence driver behaviour during automated driving [32]. While the Advanced Driving Simulator used in this study has a motion platform, the platform was stationary once the vehicle was moved into the ideal operating position. Similarly, steering wheel motions may have a similar affect and were also not simulated in this study. These limitations may have been mitigated by the fact that our study only analysed highway driving, during which such acceleration and motion cues are less pronounced.

CONCLUSION

This paper makes a contribution in that it investigates how people may interact with an automated car (L3/4) while engaging in NDRTs that are common everyday tasks. Specifically it provides insights into driver's discretionary self-interruption behaviour during automated driving on a highway using two different display devices. It shows significant differences in the frequency and duration of such interruptions, suggesting a higher likelihood of a higher situation awareness when NDRTs are consumed over a HUD rather than over mobiles. To the best of our knowledge, it is also the first paper that quantifies the self-interruption behaviour in automated driving. This behaviour not only has implications for a driver's fallback-readiness, but also for future in-vehicle systems and Human-Machine Interfaces (HMI) that aim to assess or predict, and improve the driver's fallback-readiness.

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